```
#define TRIG_PIN 9
#define TRIG_PIN2 8
#define ECHO_PIN 10
#define ECHO_PIN2 12
#define MOTOR_IN1 6
#define MOTOR_IN2 7
#define MOTOR_IN3 5
#define MOTOR_IN4 4

long duration;
int distance;
long duration2;
int distance2;

bool movingForward = true;  // Keeps track of the motor's direction

void setup() {
  pinMode(TRIG_PIN, OUTPUT);
  pinMode(TRIG_PIN2, OUTPUT);

  pinMode(ECHO_PIN, INPUT);
  pinMode(ECHO_PIN2, INPUT);

  pinMode(MOTOR_IN1, OUTPUT);
  pinMode(MOTOR_IN2, OUTPUT);

  pinMode(MOTOR_IN3, OUTPUT);
  pinMode(MOTOR_IN4, OUTPUT);

  Serial.begin(9600);
}

void loop() {
  // Read distance from first sensor
  distance = getDistance(TRIG_PIN, ECHO_PIN);

  // Read distance from second sensor
  distance2 = getDistance(TRIG_PIN2, ECHO_PIN2);

  Serial.print("Distance: ");
  Serial.println(distance);
  Serial.print("Distance2: ");
  Serial.println(distance2);

  if (movingForward) {
    if (distance <= 10) {
      movingForward = false;
    } else {

```

```
      moveForward();
    }
  } else {
    if (distance2 <= 10) {
      movingForward = true;
    } else {
      moveBackward();
    }
  }

  delay(100);
}

int getDistance(int trigPin, int echoPin) {
  long duration;
  int distance;

  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  duration = pulseIn(echoPin, HIGH);
  distance = duration * 0.034 / 2;

  return distance;
}

void moveForward() {
  digitalWrite(MOTOR_IN1, HIGH);
  digitalWrite(MOTOR_IN2, LOW);
  digitalWrite(MOTOR_IN3, HIGH);
  digitalWrite(MOTOR_IN4, LOW);
}

void moveBackward() {
  digitalWrite(MOTOR_IN1, LOW);
  digitalWrite(MOTOR_IN2, HIGH);
  digitalWrite(MOTOR_IN3, LOW);
  digitalWrite(MOTOR_IN4, HIGH);
}

```